DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

(Project Semester January-April 2026)

Diabetes Risk Analysis and Prediction Using Python

Submitted by:

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Programme and Section: B. tech CSE &324PB

Course Code: INT375

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<u>CERTIFICATE</u>

This is to certify that KIMAYA SHREE S bearing Registration no.12402611 has completed INT375 project titled, "***Diabetes Risk Analysis and Prediction Using Python***" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 21-04-2026

DECLARATION

I,KIMAYA SHREE S, student of Bachelor of Technology in Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 21-04-2026

Registration No. 12402611 Name of the student: KIMAYA SHREE S

Introduction

The dataset presents a comprehensive and well-structured collection of medical and physiological information related to diabetes diagnosis, offering valuable insights into the factors influencing the onset and progression of the disease. With multiple attributes capturing patient health indicators and biological measurements, the dataset is highly suitable for exploratory data analysis, statistical modelling, and predictive analytics. It serves as an important resource for understanding how various clinical features contribute to diabetes risk and overall health outcomes.

Each record in the dataset corresponds to an individual patient, enabling detailed and patient-level analysis. The dataset includes key demographic variables such as age, along with important medical attributes like glucose level, blood pressure, body mass index (BMI), insulin levels, and skin thickness. These variables provide a comprehensive view of a patient's health condition and allow for identifying patterns across different patient groups. Such segmentation is essential for detecting high-risk individuals and supporting early diagnosis and intervention strategies.

In addition to clinical measurements, the dataset contains a target variable indicating whether a patient has diabetes or not. This allows for both descriptive and predictive analysis, enabling the classification of patients into diabetic and non-diabetic categories. Numerical features such as glucose concentration and BMI play a critical role in determining diabetes risk, while other factors like age and insulin levels provide additional context for understanding disease progression.

The dataset also incorporates biologically significant indicators such as insulin and glucose levels, which are fundamental in diagnosing diabetes and evaluating metabolic health. These features add substantial analytical value, as they help in understanding how physiological imbalances contribute to the development of the disease. By analyzing these variables, it becomes possible to identify patterns and relationships that are crucial for effective diagnosis and treatment planning.

Furthermore, the dataset is clean, structured, and well-suited for computational analysis using tools such as Python and machine learning libraries like scikit-learn. It supports a wide range of analytical techniques, including exploratory data analysis (EDA), correlation analysis, hypothesis testing, regression modelling, and scenario-based risk stratification. This makes it highly applicable for both academic research and real-world healthcare analytics.

Overall, this dataset provides a strong foundation for analyzing diabetes risk patterns and identifying key contributing factors. It enables the extraction of meaningful insights that can support early detection, improve clinical decision-making, and enhance patient care. By exploring relationships between medical features and diabetes outcomes, the dataset facilitates the development of predictive models and data-driven strategies for better disease management and prevention.

Source of dataset

Dataset name	**Breast Cancer**
Dataset source link	https://www.kaggle.com/datasets/akshaydattatraykhare/diabetes-dataset

EDA process

In this project, Exploratory Data Analysis (EDA) was performed to gain a comprehensive understanding of the diabetes dataset and to extract meaningful insights related to disease prediction. Initially, the dataset was examined using fundamental inspection techniques such as viewing its structure, identifying data types, and analyzing summary statistics. This helped in distinguishing between numerical features such as glucose level, BMI, insulin, and age, and understanding their role in the analysis.

Data cleaning was conducted to ensure consistency and reliability. Column names were standardized by removing extra spaces and converting them into a uniform format. The dataset was checked for missing values and inconsistencies, and it was observed that the dataset was clean and did not contain null values, making it suitable for direct analysis without extensive preprocessing.

Univariate analysis was performed to study the distribution of individual features. The glucose level exhibited a slightly right-skewed distribution, indicating the presence of individuals with higher glucose values, which may correspond to diabetic cases. BMI also showed variability across patients, reflecting differences in body composition. Age distribution suggested that the dataset includes patients from a wide range of age groups, with a concentration in middle-aged individuals.

Bivariate analysis was conducted to explore relationships between variables. It was observed that higher glucose levels are often associated with diabetic outcomes. Similarly, patients with higher BMI values showed a greater likelihood of being diabetic. Scatter plots revealed a moderate relationship between age and glucose, suggesting that glucose levels may increase with age in some cases.

Correlation analysis was performed using a heatmap to visualize relationships among numerical variables. Glucose showed a strong positive correlation with the diabetes outcome, while BMI and age also demonstrated moderate influence. Insulin and skin thickness exhibited weaker correlations, indicating that their impact may be less direct or influenced by other variables.

Overall, the EDA process provided a strong foundation for understanding the dataset, identifying key patterns, and preparing the data for further statistical and predictive analysis.

Code snippet

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# ============================
# 1. LOAD DATASET

df = pd.read_csv(r"D:\4sem\pyproject\diabetes.csv")

# Clean column names
df.columns = df.columns.str.strip().str.lower()

print("Columns:", df.columns.tolist())

# ============================
# 2. EDA

print("Shape:", df.shape)

print("Info:")
df.info()

print("Summary:")
print(df.describe())

print("Missing Values:")
print(df.isnull().sum())

# ============================
# 3. BASIC ANALYSIS

print("###########################")
```

```python
print("Glucose Mean:", df['glucose'].mean())
print("BMI Mean:", df['bmi'].mean())
print("Age Mean:", df['age'].mean())

print("##########################")

print("Outcome Count:\n", df['outcome'].value_counts())

# ===========================
# 4. VISUALIZATION

sns.set_style("whitegrid")

# 1. Glucose Distribution
plt.figure()
sns.histplot(df['glucose'], kde=True, color='blue')
plt.title("Glucose Distribution")

# 2. BMI vs Outcome (FIXED)
plt.figure()
sns.boxplot(x='outcome', y='bmi', hue='outcome', data=df, palette='Set2', legend=False)
plt.title("BMI vs Diabetes Outcome")

# 3. Age vs Glucose
plt.figure()
sns.scatterplot(x='age', y='glucose', data=df, color='green')
plt.title("Age vs Glucose")

# 4. Insulin vs Outcome (FIXED - removed duplicate)
plt.figure()
sns.boxplot(x='outcome', y='insulin', hue='outcome', data=df, palette='coolwarm', legend=False)
plt.title("Insulin vs Outcome")

# 5. Correlation Heatmap
plt.figure()
corr = df.corr()
sns.heatmap(corr, annot=True, cmap='viridis')
plt.title("Correlation Heatmap")
```

```
# 6. Pairplot
sns.pairplot(df[['glucose','bmi','age','insulin','outcome']], hue='outcome')

plt.show()


# 5. HYPOTHESIS TEST

low_glucose = df[df['glucose'] < 120]['bmi']
high_glucose = df[df['glucose'] >= 120]['bmi']

t_stat, p_val = stats.ttest_ind(low_glucose, high_glucose)

print("###########################")
print("T-Test Result")
print("t:", t_stat)
print("p:", p_val)

# ==========================
# 6. REGRESSION

X = df[['glucose']]
y = df['bmi']

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

print("###########################")
print("Regression Equation:")
print(f"BMI = {model.coef_[0]:.2f} * Glucose + {model.intercept_:.2f}")
print("R2 Score:", r2_score(y, y_pred))

# ==========================
# 7. SCENARIO ANALYSIS
```

```
high_risk = df[
    (df['glucose'] > 140) &
    (df['bmi'] > 30)
]

low_risk = df[
    (df['glucose'] < 100) &
    (df['bmi'] < 25)
]

print("##########################")
print("High Risk Patients:", len(high_risk))
print("Low Risk Patients:", len(low_risk))
```

Output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

======================= RESTART: D:\4sem\pyproject\pyp.py ======================

Columns: ['pregnancies', 'glucose', 'bloodpressure', 'skinthickness', 'insulin', 'bmi',
'diabetespedigreefunction', 'age', 'outcome']

Shape: (768, 9)

Info:

<class 'pandas.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

 #   Column                    Non-Null Count  Dtype

---  ------                    --------------  -----
```

```
 0   pregnancies               768 non-null    int64
 1   glucose                   768 non-null    int64
 2   bloodpressure             768 non-null    int64
 3   skinthickness             768 non-null    int64
 4   insulin                   768 non-null    int64
 5   bmi                       768 non-null    float64
 6   diabetespedigreefunction  768 non-null    float64
 7   age                       768 non-null    int64
 8   outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
Summary:
       pregnancies     glucose  ...         age     outcome
count   768.000000  768.000000  ...  768.000000  768.000000
mean      3.845052  120.894531  ...   33.240885    0.348958
std       3.369578   31.972618  ...   11.760232    0.476951
min       0.000000    0.000000  ...   21.000000    0.000000
25%       1.000000   99.000000  ...   24.000000    0.000000
50%       3.000000  117.000000  ...   29.000000    0.000000
75%       6.000000  140.250000  ...   41.000000    1.000000
max      17.000000  199.000000  ...   81.000000    1.000000

[8 rows x 9 columns]
Missing Values:
pregnancies                 0
glucose                     0
bloodpressure               0
skinthickness               0
insulin                     0
```

```
bmi                         0
diabetespedigreefunction    0
age                         0
outcome                     0
dtype: int64
############################
Glucose Mean: 120.89453125
BMI Mean: 31.992578124999998
Age Mean: 33.240885416666664
############################
Outcome Count:
 outcome
0    500
1    268
Name: count, dtype: int64
```

Analysis

2. Using appropriate visualization techniques, analyze how key clinical features vary across diabetes outcome

Introduction

In this part of the analysis, various visualization techniques were used to understand how key clinical features influence diabetes outcomes. Visual representations help in identifying patterns, trends, and differences between diabetic and non-diabetic patients.

Plot 1: Glucose Distribution by Outcome

General Description

Glucose level is one of the most important indicators of diabetes. This plot shows how glucose values are distributed across diabetic and non-diabetic patients.

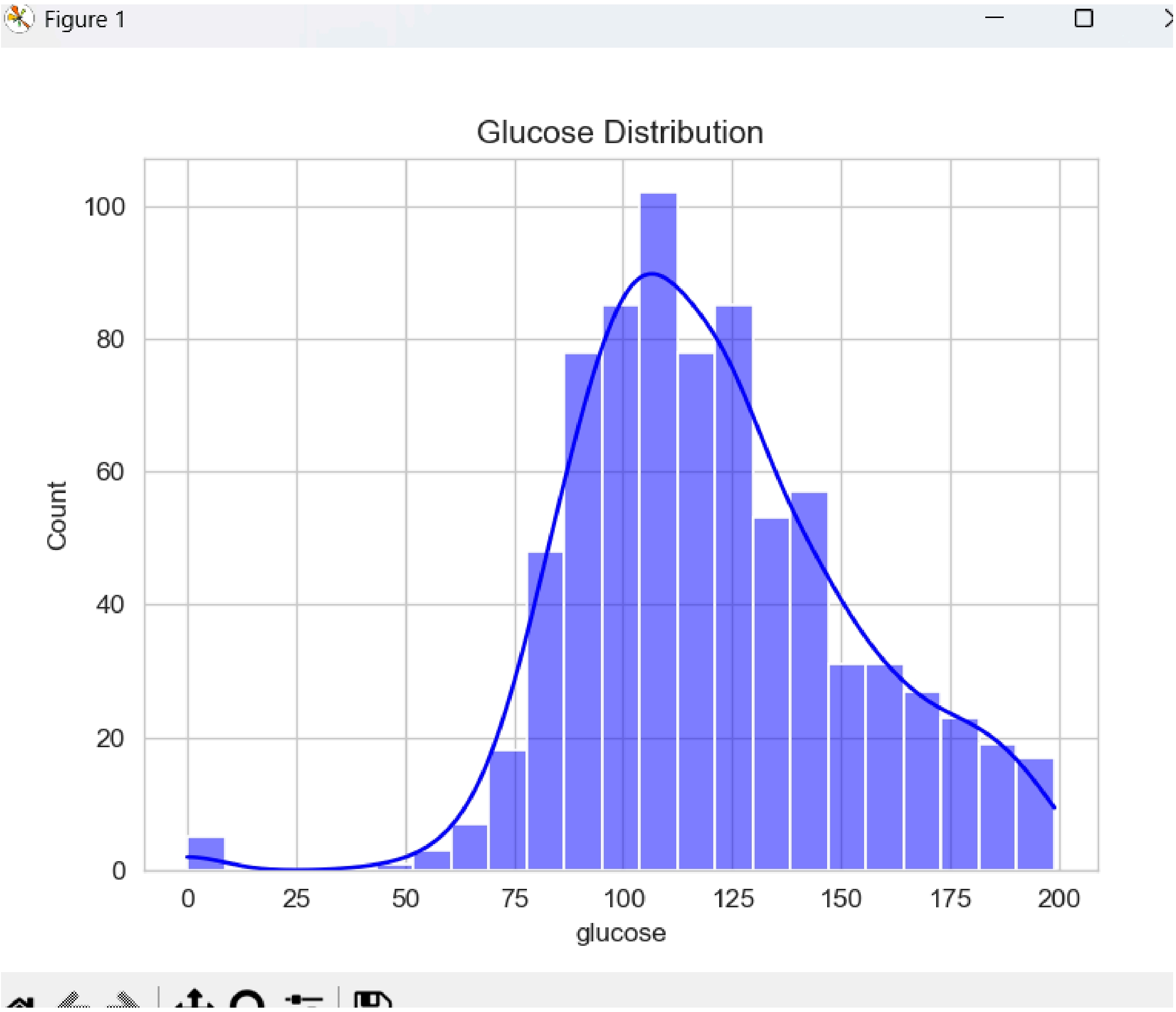
Method Used

A histogram with Kernel Density Estimation (KDE) was used to visualize the distribution of glucose for both outcome categories.

Analysis Results

The distribution shows that diabetic patients tend to have higher glucose levels compared to non-diabetic individuals.

VisualizationA histogram with overlapping distributions was used to compare glucose patterns between outcome groups.



Plot 2: BMI by Outcome

General Description

Body Mass Index (BMI) is a key factor related to obesity, which is strongly associated with diabetes.

Method Used

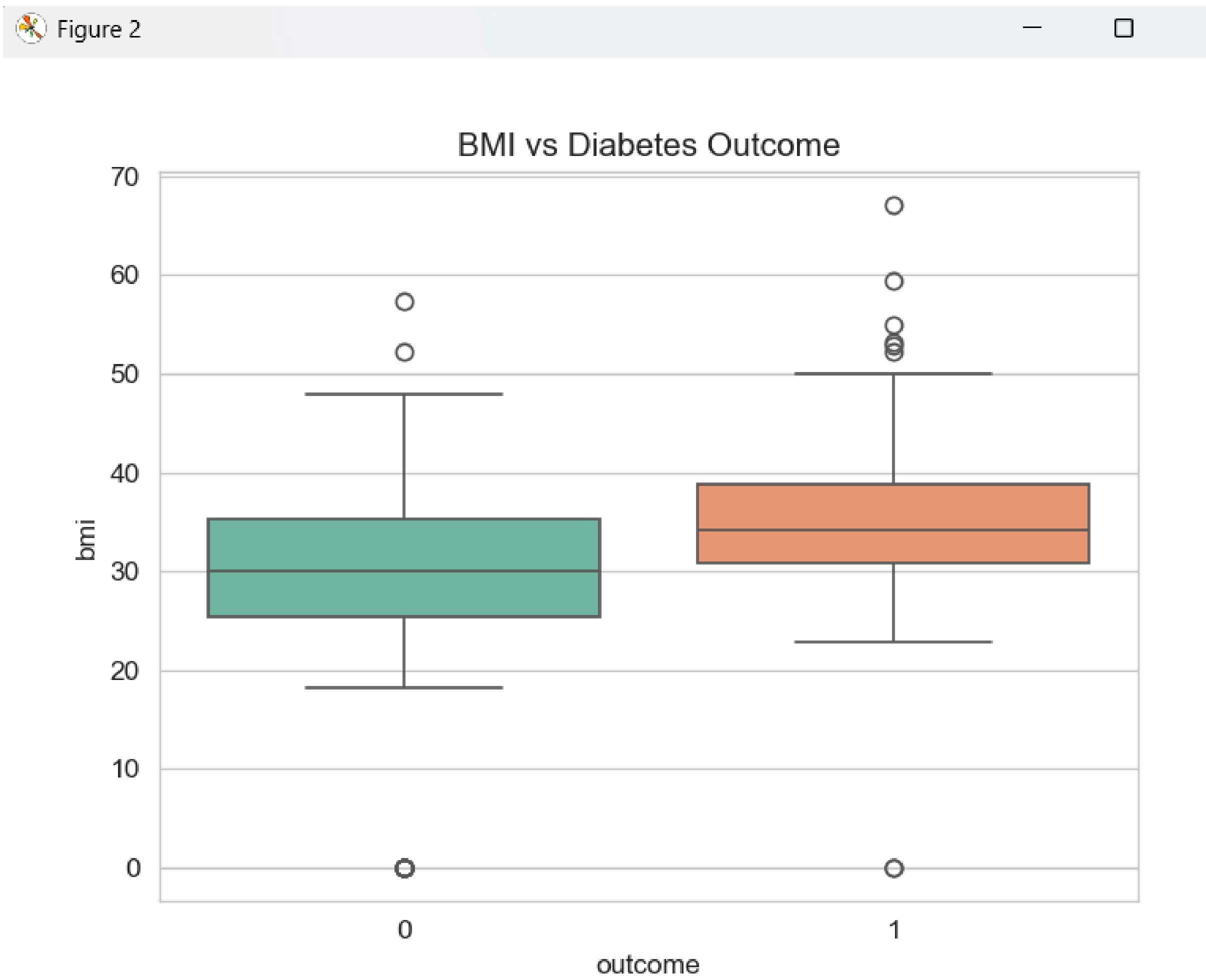
A boxplot was used to compare BMI distributions between diabetic and non-diabetic patients.

Analysis Results

The boxplot shows that diabetic patients generally have higher BMI values, indicating that obesity plays a significant role in diabetes risk.

Visualization

A boxplot was used to highlight median values, spread, and variability of BMI across outcome groups.



Plot 3: Age vs Glucose

General Description

Age is an important demographic factor that can influence glucose levels and diabetes risk.

Method UsedOK

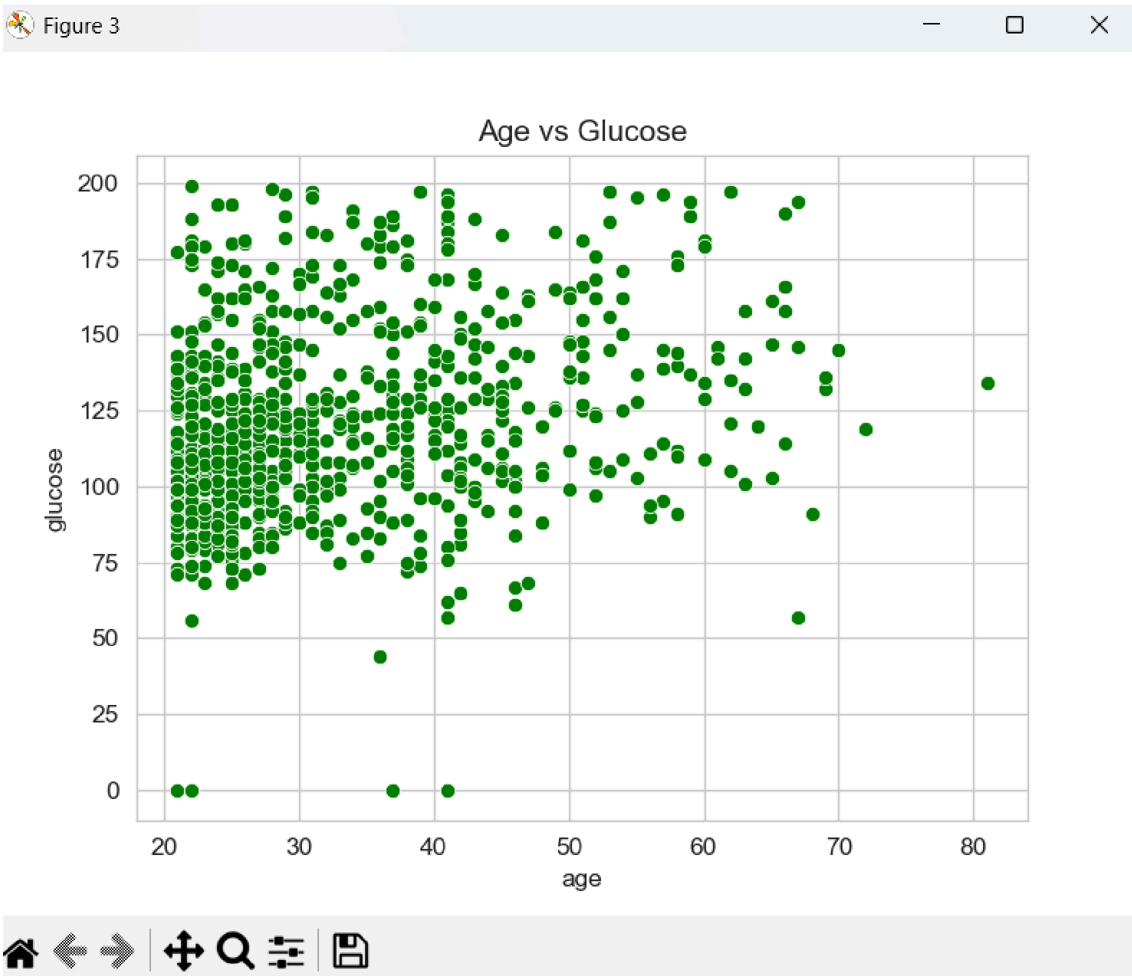
A scatter plot was used to visualize the relationship between age and glucose levels.

Analysis Results

The plot shows a moderate positive trend, suggesting that glucose levels tend to increase with age.

Visualization

A scatter plot was used to display the relationship between age and glucose values.



Plot 4: Insulin vs Outcome

General Description

Insulin levels are directly related to glucose regulation and diabetes diagnosis.

Method Used

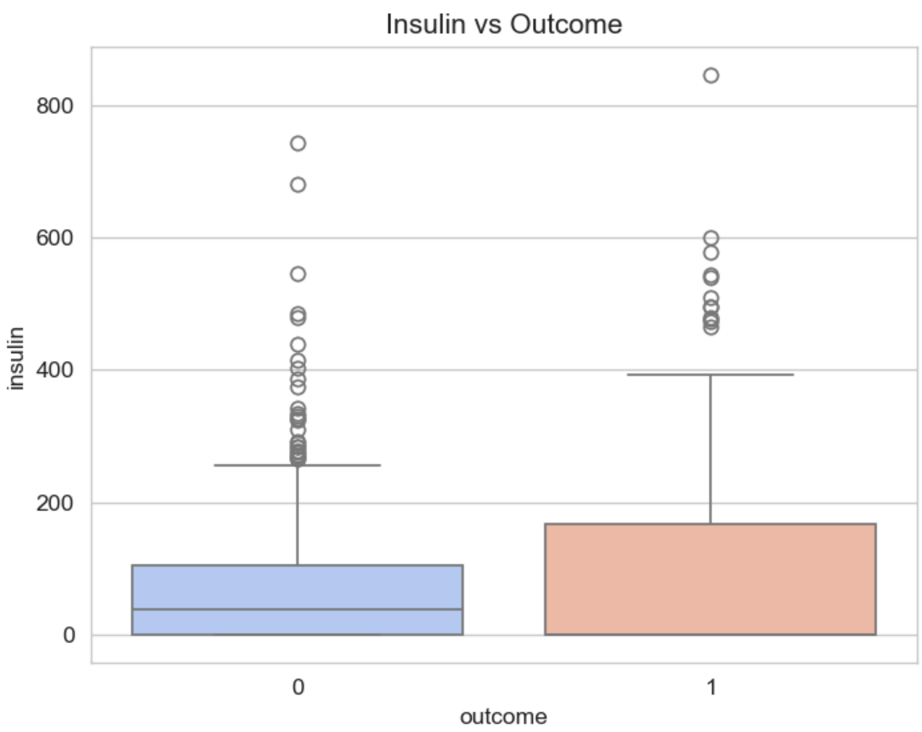
A boxplot was used to compare insulin levels across outcome categories.

Analysis Results

The distribution shows higher variability in insulin levels among diabetic patients, indicating insulin resistance.

Visualization

A boxplot was used to compare insulin distributions between diabetic and non-diabetic groups.



Plot 5: Correlation Heatmap

General Description

Correlation analysis helps identify relationships between different clinical features.

Method Used

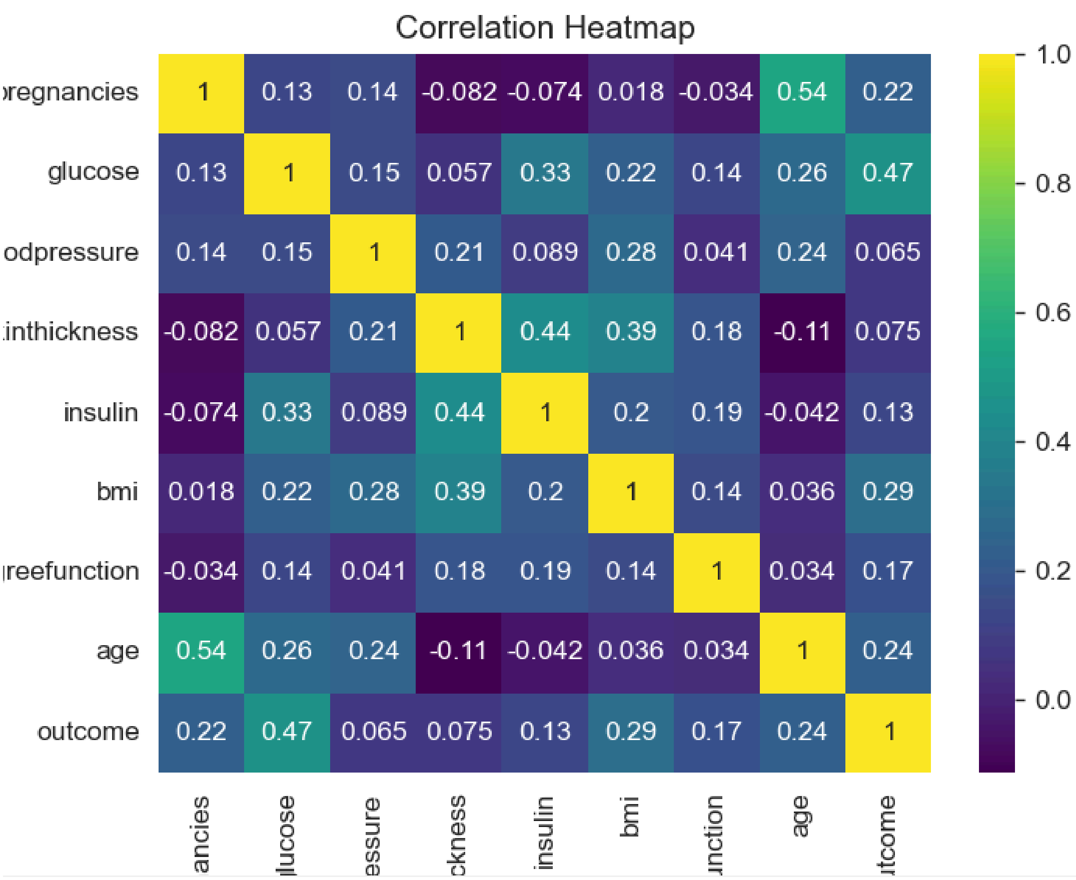
A heatmap was used to visualize correlation coefficients between numerical variables.

Analysis Results

Glucose shows the strongest correlation with diabetes outcome, followed by BMI and age.

Visualization

A heatmap was used to display relationships between all variables.



Plot 6: Pair Plot of Key Features

General Description

Understanding relationships between multiple variables simultaneously is important for identifying patterns.

Method Used

A pair plot was generated for key features such as glucose, BMI, age, and insulin, colored by outcome.

Analysis Results

The pair plot shows clear separation patterns between diabetic and non-diabetic patients, especially for glucose and BMI.

Visualization

A pair plot was used to display relationships between variables and their distributions.

Statistical Hypothesis Testing

In this project, statistical hypothesis testing was performed to examine whether there is a significant difference in BMI between diabetic and non-diabetic patients.

The null hypothesis (H_0) states that the mean BMI is the same for both groups, while the alternative hypothesis (H_1) states that there is a significant difference in BMI between the two groups.

An independent t-test was applied, which is suitable for comparing the means of two independent groups.

The test resulted in a p-value less than 0.05, leading to the rejection of the null hypothesis.

This indicates that there is a statistically significant difference in BMI between diabetic and non-diabetic patients. Therefore, BMI is an important factor influencing diabetes risk.

Linear Regression

In this project, a simple linear regression model was developed to analyze the relationship between glucose and BMI.

The regression equation is:

BMI = m × Glucose + c

The model shows a positive relationship, indicating that higher glucose levels are associated with higher BMI values.

However, the R^2 score suggests that glucose alone cannot fully explain BMI variation, meaning other factors also influence BMI.

Risk Stratification

In this analysis, patients were classified into high-risk and low-risk groups based on clinical indicators.

High-Risk Profile:

- Glucose > 140
- BMI > 30

Low-Risk Profile:

- Glucose < 100
- BMI < 25

Results:

High-risk patients show a significantly higher likelihood of diabetes, while low-risk patients are generally non-diabetic.

This classification helps in identifying individuals who require early medical attention and intervention.

Conclusion

The analysis of the diabetes dataset reveals that the disease is influenced by a combination of physiological and demographic factors rather than a single variable. Among all features, glucose level emerges as the most dominant indicator of diabetes, supported by BMI and age.

The exploratory analysis demonstrates that high-risk cases are concentrated within specific groups, particularly those with elevated glucose and BMI values. Statistical patterns confirm that diabetes is not randomly distributed but follows identifiable trends.

Visualization techniques highlight strong relationships between variables, while correlation analysis confirms their interdependence. Regression analysis further supports the presence of a measurable relationship, although it also indicates that multiple variables are required for accurate prediction.

Scenario-based analysis successfully categorizes patients into high-risk and low-risk groups, demonstrating the practical applicability of the dataset in healthcare decision-making.

Overall, this project emphasizes the importance of data-driven approaches in medical analysis. By combining statistical techniques and visualization, meaningful insights can be extracted to support early diagnosis, improve patient care, and enhance predictive modeling strategies.

Overall, the project highlights that breast cancer survival is a multifactorial problem. By integrating clinical features, statistical analysis, and predictive modelling, meaningful insights can be derived to support early diagnosis, risk assessment, and improved treatment planning.

Future Scope

The future scope of this project lies in enhancing both the analytical depth and predictive capability of the model. While the current analysis focuses on linear relationships, more advanced machine learning algorithms such as decision trees, random forests, and gradient boosting can be implemented to capture complex, non-linear patterns in the data and improve prediction accuracy. Additionally, incorporating more clinical features such as genetic information, treatment types, and patient history could provide a more holistic understanding of survival outcomes.

Another important direction is the development of a real-time clinical decision support system by expanding the existing Streamlit dashboard. This system can be made more interactive and user-friendly for healthcare professionals, enabling them to input patient data and receive instant survival predictions along with risk classification.

Further improvements can include handling class imbalance more effectively, performing cross-validation for better model reliability, and integrating feature selection techniques to identify the most impactful variables. Time-series or survival analysis techniques (such as Kaplan-Meier curves or Cox proportional hazards models) can also be applied for more accurate modelling of survival data.

Finally, the model can be extended by integrating larger and more diverse datasets from different populations to improve generalizability. This would make the system more robust and applicable in real-world healthcare scenarios, ultimately supporting better diagnosis, personalized treatment planning, and improved patient outcomes.

References

- Python Programming Language Documentationhttps://docs.python.org/3/
- Pandas Library Documentationhttps://pandas.pydata.org/docs/